

QUANTUM COMPUTING FOUNDATIONS

Mathematical and Physical Foundations for Quantum Computing

A rigorous foundation for qubits, circuits, Hamiltonians, and Ising models

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CyberColombia · University text with worked examples and exercises

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Mathematical and Physical Foundations for Quantum Computing

A university introduction to the mathematical and physical prerequisites for quantum computing, with worked examples, exercises, reasoned solutions, and reproducible computational laboratories.

Fourth revised edition, July 2026, version 09. English edition based on Spanish version 09.

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Purpose and scope

This book introduces the mathematical and physical prerequisites for a university course in quantum computing. It does not assume that readers already know the field's notation. Each symbol and structure is introduced first in ordinary language, then through examples, and finally with the formal detail needed to use it correctly.

The chapter sequence follows concrete conceptual dependencies. A quantum state cannot be defined clearly before vectors and complex numbers; measurement probabilities require inner products and amplitudes; observables require matrices, eigenvalues, and operators; multi-qubit circuits require tensor products; and time evolution requires unitary operators and Hamiltonians.

Learning objectives

- Build the mathematical language used for quantum states and operators.
- Connect exact probabilities with finite experimental counts.
- Develop qubits, density matrices, gates, circuits, and entanglement.
- Relate classical and quantum Hamiltonian dynamics.
- Verify central results with reproducible Python laboratories.

Scope

The complete theory of operators and Hilbert spaces requires specialized courses in functional analysis. This text develops the conceptual and mathematical core needed to begin quantum mechanics and quantum computing on a sound basis, while identifying the points at which more advanced treatment is required.

How to use this book

Definitions and central results appear in teal; worked examples in green; warnings and common errors in amber or red. Each conceptual chapter provides worked examples and four exercises: analytical, conceptual, diagnostic, and computational. The laboratories use Python to verify the mathematics, and the solutions appendix explains both the correct reasoning and frequent mistakes.

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PART I

Mathematical language and quantum states

1 Mathematical prerequisites

Learning objectives

- Use sets, functions, scalars, vectors, matrices, and linear maps precisely.
- Work with bases, inner products, orthogonality, eigenvalues, and tensor products.
- Distinguish an abstract vector from its coordinates in a selected basis.

1.1 Sets, elements, and functions

A **set** is a collection of identified objects called elements. We write $x \in A$ when x belongs to A , $x \notin A$ otherwise, and $A \subseteq B$ when every element of A also belongs to B . A function

$$f : A \longrightarrow B$$

assigns exactly one output $f(x) \in B$ to every input $x \in A$. The set A is the domain and B is the codomain.

Example 1.1 · A finite function

Let $A = \{-3, -2, -1, 0, 1, 2, 3\}$ and $B = \{0, 1, 4, 9\}$. The rule $f(x) = x^2$ defines a function $f : A \rightarrow B$; for example, $f(-3) = 9$.

1.2 Number systems and fields

The main number systems are

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

They are, respectively, the natural numbers, integers, rational numbers, real numbers, and complex numbers. In this book $0 \in \mathbb{N}$. A **field** $\mathbb{F}$ is a set in which addition, subtraction, multiplication, and division by a nonzero element are closed and satisfy the usual algebraic laws. The fields used here are $\mathbb{R}$ and $\mathbb{C}$.

1.3 Scalars, vectors, and vector spaces

A **scalar** is an element of $\mathbb{F}$. A **vector** belongs to a vector space V over $\mathbb{F}$. Vectors can be columns, functions, polynomials, or matrices. Their role is determined by the rules for vector addition and scalar multiplication, rather than by their visual form.

For every $u, v, w \in V$ and $a, b \in \mathbb{F}$, a vector space satisfies the associative and commutative laws for addition, has a zero vector and additive inverses, and obeys

$$\begin{aligned} a(u + v) &= au + av, & (a + b)v &= av + bv, \\ a(bv) &= (ab)v, & 1v &= v. \end{aligned}$$

Example 1.2 · Geometric vectors

In $\mathbb{R}^2$, let $u = (1, 2)^T$ and $v = (-1, 3)^T$. Then

$$2u - v = (2, 4)^T - (-1, 3)^T = (3, 1)^T.$$

The number 2 is a scalar; u , v , and $2u - v$ are vectors.

Example 1.3 · Functions as vectors

The polynomials $P_2 = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}$ form a vector space. For example,

$$2(1 + x) - (3 - x^2) = -1 + 2x + x^2 \in P_2.$$

1.4 Linear combinations, independence, and bases

A linear combination of $v_1, \dots, v_k$ is $\alpha_1 v_1 + \dots + \alpha_k v_k$. Their span contains every such combination. The vectors are linearly independent if

$$\alpha_1 v_1 + \dots + \alpha_k v_k = 0 \implies \alpha_1 = \dots = \alpha_k = 0.$$

A **basis** is an independent set that spans the whole space; the number of basis vectors is the dimension.

Example 1.4 · A change of coordinates

The vectors $b_1 = (1, 1)^T$ and $b_2 = (1, -1)^T$ form a basis of $\mathbb{R}^2$. For $v = (3, 1)^T$, solving $v = c_1 b_1 + c_2 b_2$ gives $c_1 = 2$ and $c_2 = 1$. The vector did not change; its coordinates changed with the basis.

1.5 Matrices and linear transformations

A matrix $A = (A_{ij}) \in \mathbb{F}^{m \times n}$ represents a linear map after bases have been chosen. Linearity means

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v).$$

The kernel is $\ker T = \{v : Tv = 0\}$; the image contains all possible outputs; the rank is the dimension of the image. In finite dimensions,

$$\dim \ker T + \text{rank } T = \dim V.$$

For a matrix A , A^T is the transpose, A^* the entrywise conjugate, $A^\dagger = (A^*)^T$ the adjoint, A^{-1} the inverse when it exists, $\text{tr } A$ the trace, and $\det A$ the determinant. Products reverse under transpose and adjoint:

$$(AB)^T = B^T A^T, \quad (AB)^\dagger = B^\dagger A^\dagger.$$

1.6 Inner products, norms, and orthogonality

On $\mathbb{C}^n$ the standard inner product is

$$\langle u | v \rangle = u^\dagger v = \sum_{k=1}^n u_k^* v_k, \quad \|v\| = \sqrt{\langle v | v \rangle}.$$

Vectors are orthogonal when $\langle u | v \rangle = 0$. An orthonormal set has unit vectors that are pairwise orthogonal.

The Gram-Schmidt procedure converts independent vectors into an orthonormal set. Given $v_1, \dots, v_k$, define

$$w_j = v_j - \sum_{i=1}^{j-1} \langle u_i | v_j \rangle u_i, \quad u_j = \frac{w_j}{\|w_j\|}.$$

Complex inner products

With the physics convention used here, the first argument is conjugate-linear and the second is linear. Omitting the conjugate can produce nonreal norms and invalid probabilities.

1.7 Eigenvalues and diagonalization

A nonzero vector v is an eigenvector of A with eigenvalue λ if $Av = \lambda v$. Eigenvalues satisfy $\det(A - \lambda I) = 0$. A matrix is diagonalizable when $A = SDS^{-1}$ for an invertible S and diagonal D .

Example 1.5 • A simple spectrum

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, the eigenvalues are 3 and 1 with normalized eigenvectors $2^{-1/2}(1, 1)^T$ and $2^{-1/2}(1, -1)^T$. Their orthogonality anticipates the spectral theorem for Hermitian matrices.

1.8 Tensor products

The tensor product builds composite systems. For $u \in \mathbb{C}^m$ and $v \in \mathbb{C}^n$, $u \otimes v \in \mathbb{C}^{mn}$. If $u = (u_1, u_2)^T$ and $v = (v_1, v_2)^T$, then

$$u \otimes v = (u_1 v_1, u_1 v_2, u_2 v_1, u_2 v_2)^T.$$

For matrices $A = (a_{ij})$, the product $A \otimes B$ is the block matrix whose (i, j) block is $a_{ij}B$. It satisfies

$$(A \otimes B)(u \otimes v) = (Au) \otimes (Bv).$$

1.9 Exercises

Exercise 1.1

Determine whether $(1, i)^T$ and $(i, 1)^T$ are orthogonal in $\mathbb{C}^2$.

Exercise 1.2

Apply Gram-Schmidt to $(1, 1)^T$ and $(1, i)^T$ in $\mathbb{C}^2$.

Exercise 1.3

Explain why a vector and its coordinate column should not be identified without specifying a basis.

Exercise 1.4

Use NumPy to verify the eigenpairs of $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and compute $A \otimes I_2$.

References and further reading

Axler, S. (2024). *Linear algebra done right* (4th ed.). Springer.

Strang, G. (2016). *Introduction to linear algebra* (5th ed.). Wellesley-Cambridge Press.

2 Complex numbers

Learning objectives

- Use Cartesian, polar, and exponential forms of a complex number.
- Compute conjugates, moduli, phases, and adjoints.
- Distinguish global phase from observable relative phase.

2.1 Definition and operations

A complex number has the form $z = a + ib$, with $a, b \in \mathbb{R}$ and $i^2 = -1$. The real and imaginary parts are $\text{Re } z = a$ and $\text{Im } z = b$. Addition is componentwise, while multiplication uses $i^2 = -1$:

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc).$$

Example 2.1 · Multiplication

$(2+3i)(1-i) = 5+i$. The cross terms contribute $-2i+3i = i$, and $3i(-i) = 3$.

2.2 Conjugate and modulus

The conjugate of $z = a + ib$ is $z^* = a - ib$. Its modulus is

$$|z| = \sqrt{z^* z} = \sqrt{a^2 + b^2}.$$

Conjugation reverses the sign of the phase and distributes over sums and products. For $z \neq 0$, $z^{-1} = z^* / |z|^2$.

Example 2.2 · Division

$$\frac{1+i}{2-i} = \frac{(1+i)(2+i)}{(2-i)(2+i)} = \frac{1+3i}{5}.$$

2.3 Polar form and Euler's formula

Euler's identity $e^{i\theta} = \cos \theta + i \sin \theta$ gives

$$z = re^{i\theta}, \quad r = |z|, \quad \theta = \arg z.$$

Multiplication multiplies moduli and adds phases. In particular, $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$.

Example 2.3 · Roots of unity

The solutions of $z^4 = 1$ are $e^{ik\pi/2}$ for $k = 0, 1, 2, 3$, namely $1, i, -1, -i$. They lie uniformly on the unit circle.

2.4 Complex vectors and adjoints

If $v = (v_1, \dots, v_n)^T$, then

$$v^\dagger = (v_1^*, \dots, v_n^*), \quad v^\dagger v = \sum_k |v_k|^2.$$

The adjoint is essential because quantum amplitudes are complex while probabilities must be real and nonnegative.

Example 2.4 · Normalizing a state vector

For $v = (1, 2i)^T$, $v^\dagger v = 5$, so $\psi = 5^{-1/2}(1, 2i)^T$ is normalized.

2.5 Global and relative phase

Vectors $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ represent the same pure physical state because every probability is unchanged. This is a **global phase**. By

contrast, in

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle),$$

the angle ϕ is a relative phase and can change interference outcomes.

Phase is not always removable

A phase multiplying the entire state is unobservable. A phase multiplying only one component relative to another is generally observable after a change of measurement basis.

2.6 Exercises

Exercise 2.1

Write $-1 + i\sqrt{3}$ in polar and exponential form.

Exercise 2.2

Normalize $(1 - i, 2)^T$ and verify its norm using the adjoint.

Exercise 2.3

Explain why $|0\rangle + |1\rangle$ and $|0\rangle - |1\rangle$ do not differ only by a global phase.

Exercise 2.4

Use Python to plot the four fourth roots of unity and verify $z^4 = 1$ numerically.

References and further reading

Needham, T. (1997). *Visual complex analysis*. Oxford University Press.

Susskind, L., & Friedman, A. (2014). *Quantum mechanics: The theoretical minimum*. Basic Books.

3 Probability and statistics for quantum systems

Learning objectives

- Work with random variables, conditional probability, and Bayes' rule.
- Convert complex amplitudes into outcome probabilities.
- Quantify sampling error in finite measurement counts.

3.1 Events, random variables, and distributions

An event is a subset of possible outcomes. A random variable X assigns a number to each outcome. For a discrete distribution with values x_k and probabilities p_k ,

$$\mathbb{E}[X] = \sum_k x_k p_k, \quad \text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

Example 3.1 · A binary outcome

If $X \in \{-1, +1\}$ and $P(X = +1) = p$, then $\mathbb{E}[X] = 2p - 1$ and $\text{Var}(X) = 4p(1 - p)$.

3.2 Joint and conditional probability

For events A and B with $P(B) > 0$,

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

The events are independent when $P(A \cap B) = P(A)P(B)$. Independence is a property of a probability model; it should not be confused with orthogonality or quantum separability.

Example 3.2 · Conditional outcomes

If $P(A) = 0.4$ and $P(B | A) = 0.75$, then $P(A \cap B) = P(B | A)P(A) = 0.30$.

3.3 Bayes' rule

Bayes' rule reverses a condition:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

Example 3.3 · Rare events

Suppose a detector is positive with probability 0.95 when a condition is present, with probability 0.01 otherwise, and the prior probability is 0.001. Then

$$P(A | +) = \frac{0.95(0.001)}{0.95(0.001) + 0.01(0.999)} \approx 0.087.$$

A highly sensitive detector can still have a modest posterior probability when the event is rare.

3.4 Quantum amplitudes and probabilities

For a normalized state $|\psi\rangle = \sum_k \alpha_k |k\rangle$, the Born rule gives

$$P(k) = |\alpha_k|^2, \quad \sum_k |\alpha_k|^2 = 1.$$

Amplitudes add before the modulus is squared. This permits constructive and destructive interference.

Example 3.4 · Interference

Two indistinguishable paths with amplitudes $1/2$ and $-1/2$ have total amplitude zero and therefore zero probability, even though the separate squared moduli sum to $1/2$.

3.5 Repetitions, counts, and standard error

Exact probabilities describe an ideal model. Experiments estimate them from N repetitions, often called shots. If outcome 1 is observed n times,

$$\hat{p} = \frac{n}{N}, \quad \text{SE}(\hat{p}) \approx \sqrt{\frac{\hat{p}(1 - \hat{p})}{N}}.$$

For measured values $a_1, \dots, a_N$, the sample mean estimates the expectation:

$$\langle \widehat{A} \rangle = \frac{1}{N} \sum_{j=1}^N a_j, \quad \text{SE} \approx \frac{s_A}{\sqrt{N}}.$$

The $N^{-1/2}$ scaling means that halving the standard error requires about four times as many repetitions.

Finite-sample interpretation

An observed frequency is an estimate, not the exact Born probability. Report the number of shots and an uncertainty measure whenever empirical counts are used to support a conclusion.

3.6 Exercises

Exercise 3.1

For outcomes 0 and 1 with probabilities 0.7 and 0.3, compute the mean and variance.

Exercise 3.2

A run produces 240 ones in 1,000 shots. Estimate p , its standard error, and a 95 percent normal-approximation interval.

Exercise 3.3

Explain why adding probabilities instead of amplitudes removes interference.

Exercise 3.4

Simulate Bernoulli sampling for $N = 100, 1000, 10000$ and verify the $N^{-1/2}$ decrease in error.

References and further reading

- Blitzstein, J. K., & Hwang, J. (2019). *Introduction to probability* (2nd ed.). CRC Press.
- Nielsen, M. A., & Chuang, I. L. (2010). *Quantum computation and quantum information*. Cambridge University Press.

4 Hilbert spaces and Dirac notation

Learning objectives

- Interpret quantum states as rays in a complex Hilbert space.
- Use bras, kets, orthonormal bases, completeness, and projectors.
- Form composite spaces and recognize entangled pure states.

4.1 Definition

A **Hilbert space** $\mathcal{H}$ is an inner-product space that is complete in the norm induced by its inner product. Completeness means that each Cauchy sequence converges to an element of the same space. It is automatic in finite-dimensional spaces such as $\mathbb{C}^n$, but essential for function spaces such as $L^2([0, L])$.

A pure state is represented by a normalized vector $|\psi\rangle \in \mathcal{H}$, with $\langle\psi | \psi\rangle = 1$, up to a global phase.

Example 4.1 · The Hilbert space of a qubit

The computational basis of $\mathbb{C}^2$ is

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A general normalized qubit is $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$.

Example 4.2 · Square-integrable wave functions

For a particle in $[0, L]$, the state space is $L^2([0, L])$ with

$$\langle \phi | \psi \rangle = \int_0^L \phi^*(x) \psi(x) dx.$$

The functions $\sqrt{2/L} \sin(n\pi x/L)$ form an orthonormal family.

4.2 Bra-ket notation

A vector is written as a ket $|\psi\rangle$. Its dual is the bra $\langle\psi| = (|\psi\rangle)^\dagger$. The scalar $\langle\phi | \psi\rangle$ is an inner product, while $|\psi\rangle\langle\phi|$ is an operator. In an orthonormal basis $\{|n\rangle\}$,

$$|\psi\rangle = \sum_n c_n |n\rangle, \quad c_n = \langle n | \psi \rangle.$$

Example 4.3 · Outer products

For $|+\rangle = 2^{-1/2}(|0\rangle + |1\rangle)$,

$$|+\rangle\langle+| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

This matrix is the projector onto the one-dimensional subspace spanned by $|+\rangle$.

4.3 Orthonormal bases and completeness

An orthonormal basis satisfies $\langle m | n \rangle = \delta_{mn}$ and the completeness relation

$$\sum_n |n\rangle\langle n| = I.$$

In a continuous position basis, the analogous expression is $\int |x\rangle\langle x| dx = I$, interpreted with the appropriate mathematical care.

4.4 Projectors

An orthogonal projector obeys $P^2 = P$ and $P^\dagger = P$. For a normalized vector, $P_\psi = |\psi\rangle\langle\psi|$. Projectors represent yes-no questions and subspaces.

Example 4.4 · Projection probability

For normalized $|\psi\rangle$ and a projector P , the probability of the outcome associated with P is $\langle\psi|P|\psi\rangle$. For $P_0 = |0\rangle\langle 0|$ and $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, this is $|\alpha|^2$.

4.5 Composite systems and entanglement

If systems A and B have spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, their joint space is $\mathcal{H}_A \otimes \mathcal{H}_B$. A product state has the form $|\psi\rangle_A \otimes |\phi\rangle_B$. A state that cannot be written this way is entangled.

The Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

is entangled. Neither qubit possesses its own pure state vector even though the joint state is pure.

Dimension grows multiplicatively

The state space of n qubits has dimension 2^n . This exponential growth is a central resource and also the reason that general quantum states are expensive to simulate classically.

4.6 Exercises

Exercise 4.1

Expand $|\psi\rangle = (1, i, -1, 0)^T / \sqrt{3}$ in the standard basis and verify its norm.

Exercise 4.2

Show directly that $P = |+\rangle\langle+|$ is Hermitian and idempotent.

Exercise 4.3

Explain why the completeness relation depends on using a complete basis.

Exercise 4.4

Use NumPy to test whether $|\Phi^+\rangle$ can be represented as a rank-one 2×2 coefficient matrix.

References and further reading

Hall, B. C. (2013). *Quantum theory for mathematicians*. Springer.

Preskill, J. (n.d.). *Lecture notes for Physics 229: Quantum information and computation*. California Institute of Technology.

PART II

Operators, measurement and circuits

5 Operators, adjoints, and Hermitian matrices

Learning objectives

- Represent linear operators as matrices and compute their adjoints.
- Distinguish normal, Hermitian, unitary, positive, and projector operators.
- Use commutators to test compatibility and conservation laws.

5.1 Operators and matrix representations

A linear operator $A : \mathcal{H} \rightarrow \mathcal{H}$ satisfies $A(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha A|\psi\rangle + \beta A|\phi\rangle$. In a basis $\{|n\rangle\}$, its entries are $A_{mn} = \langle m|A|n\rangle$. Changing the basis changes the matrix, but not the abstract operator.

Example 5.1 • A bit flip

The Pauli operator $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ interchanges the computational basis states: $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$.

5.2 The adjoint and normal operators

The adjoint is defined by

$$\langle \phi | A\psi \rangle = \langle A^\dagger \phi | \psi \rangle.$$

For matrices, $A^\dagger = (A^*)^T$. A matrix is normal when $AA^\dagger = A^\dagger A$. The spectral theorem states that a finite normal matrix can be diagonalized by a unitary matrix.

Example 5.2 · A normal matrix that is not Hermitian

$A = \text{diag}(1, i)$ is normal because it commutes with its adjoint. It is not Hermitian because $A^\dagger = \text{diag}(1, -i) \neq A$.

5.3 Hermitian matrices

An operator is Hermitian when $A = A^\dagger$. Its eigenvalues are real, and eigenvectors belonging to distinct eigenvalues are orthogonal. In finite dimensions,

$$A = UDU^\dagger,$$

where U is unitary and D is real diagonal.

Example 5.3 · A complex Hermitian matrix

For $A = \begin{pmatrix} 2 & i \\ -i & 3 \end{pmatrix}$, $A^\dagger = A$. Its eigenvalues are $(5 \pm \sqrt{5})/2$, both real.

5.4 Expectation values are real

For normalized $|\psi\rangle$, the expectation of a Hermitian operator is $\langle A \rangle_\psi = \langle \psi | A | \psi \rangle$. Since

$$\langle A \rangle_\psi^* = \langle \psi | A^\dagger | \psi \rangle = \langle \psi | A | \psi \rangle,$$

the expectation is real.

5.5 Unitary, projector, and positive operators

A unitary operator satisfies $U^\dagger U = UU^\dagger = I$ and therefore preserves inner products and norms. A projector satisfies $P^2 = P =$

$P^\dagger$. A positive operator satisfies $\langle \psi | A | \psi \rangle \geq 0$ for every $|\psi\rangle$; its eigenvalues are nonnegative.

Example 5.4 · Pauli matrices

The matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are both Hermitian and unitary. Their eigenvalues are ± 1 .

5.6 Products and commutators

The commutator is $[A, B] = AB - BA$. If A and B are Hermitian, their product AB is Hermitian exactly when they commute. The Pauli matrices satisfy $[X, Y] = 2iZ$ and cyclic permutations.

Hermitian does not mean unitary

$\text{diag}(2, 3)$ is Hermitian but not unitary. Hermiticity represents real-valued observables; unitarity represents reversible norm-preserving evolution. Some operators, including the Pauli matrices, have both properties.

5.7 Exercises

Exercise 5.1

Determine whether $\begin{pmatrix} 1 & 1+i \\ 1-i & 2 \end{pmatrix}$ is Hermitian.

Exercise 5.2

Prove that every projector has eigenvalues only in $\{0, 1\}$.

Exercise 5.3

Give an example showing that the product of two Hermitian matrices need not be Hermitian.

Exercise 5.4

Use NumPy to verify the commutators of X , Y , and Z .

References and further reading

Horn, R. A., & Johnson, C. R. (2013). *Matrix analysis* (2nd ed.). Cambridge University Press.

Nielsen, M. A., & Chuang, I. L. (2010). *Quantum computation and quantum information*. Cambridge University Press.

6 Observables, measurement, and uncertainty

Learning objectives

- Apply the spectral measurement rule and state update.
- Compute expectation, variance, and finite-shot estimates.
- Represent a qubit on the Bloch sphere and use uncertainty relations.

6.1 Operational postulates

An observable is represented by a self-adjoint operator $A = \sum_a a P_a$, where the real values a are possible outcomes and P_a are orthogonal spectral projectors. In a normalized state $|\psi\rangle$,

$$P(a) = \langle\psi|P_a|\psi\rangle.$$

After an ideal projective measurement yielding a , the state becomes $P_a|\psi\rangle/\sqrt{P(a)}$ when $P(a) > 0$.

Example 6.1 • Measurement of Z

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, measuring $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ returns $+1$ with probability $|\alpha|^2$ and -1 with probability $|\beta|^2$.

6.2 Expectation and variance

The expectation and variance are

$$\langle A \rangle = \langle \psi | A | \psi \rangle, \quad (\Delta A)^2 = \langle \psi | A^2 | \psi \rangle - \langle A \rangle^2.$$

An expectation is an ensemble mean, not necessarily an allowed outcome of one measurement.

Example 6.2 · Expectation is not a single result

In $|0\rangle$, a measurement of X yields $+1$ or -1 with equal probability. Thus $\langle X \rangle = 0$ although zero is not an eigenvalue of X .

For finite shots, replace the theoretical expectation by the sample mean. Reporting uncertainty connects the operator formalism to laboratory evidence.

6.3 The qubit and the Bloch sphere

Up to global phase, every pure qubit can be written

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle.$$

Its Bloch vector is

$$\mathbf{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$

More generally,

$$\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad \|\mathbf{r}\| \leq 1.$$

Pure states lie on the sphere; mixed states lie inside it.

Example 6.3 · The state $|+\rangle$

$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ has $(\theta, \phi) = (\pi/2, 0)$ and Bloch vector $(1, 0, 0)$. It is the $+1$ eigenstate of X .

6.4 Compatibility and uncertainty

For Hermitian operators A and B ,

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|.$$

Commuting observables can possess a common eigenbasis. Noncommuting observables cannot in general be assigned simultaneously sharp values in every state.

Example 6.4 · Spin uncertainty

Because $[X, Y] = 2iZ$, one has $\Delta X \Delta Y \geq |\langle Z \rangle|$. In $|0\rangle$, $\langle Z \rangle = 1$ and $\Delta X = \Delta Y = 1$, so equality holds.

State uncertainty is not instrument error

The uncertainty relation constrains the distributions predicted by the state and the noncommuting operators. Instrument noise is a separate experimental effect that must be calibrated independently.

6.5 Exercises

Exercise 6.1

Compute $\langle Z \rangle$ and ΔZ for $|\psi\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$.

Exercise 6.2

Find the Bloch vector of $\cos(\pi/8)|0\rangle + \sin(\pi/8)|1\rangle$.

Exercise 6.3

Explain why repeated measurement immediately after an ideal projective result returns the same eigenvalue.

Exercise 6.4

Simulate 10,000 Z measurements of $\sqrt{0.3}|0\rangle + \sqrt{0.7}|1\rangle$ and compare the observed mean with the theoretical expectation.

References and further reading

- Busch, P., Lahti, P., Pellonpää, J.-P., & Ylinen, K. (2016). *Quantum measurement*. Springer.
- Sakurai, J. J., & Napolitano, J. (2021). *Modern quantum mechanics* (3rd ed.). Cambridge University Press.

7 Multiple qubits, entanglement, and density matrices

Learning objectives

- Build multi-qubit registers with tensor products.
- Test separability of pure two-qubit states.
- Use density matrices and partial traces for subsystems.

7.1 Registers and product states

The standard basis of two qubits is $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. If $|\psi\rangle_A = \alpha|0\rangle + \beta|1\rangle$ and $|\phi\rangle_B = \gamma|0\rangle + \delta|1\rangle$, then

$$|\psi\rangle_A \otimes |\phi\rangle_B = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle.$$

Example 7.1 · A product register

$|+\rangle \otimes |0\rangle = (|00\rangle + |10\rangle)/\sqrt{2}$. Measurement always gives zero on the second qubit and a uniform result on the first.

7.2 Separability of pure two-qubit states

For the pure state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle,$$

form the coefficient matrix $C = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The state is separable exactly when $\text{rank } C = 1$, equivalently $ad - bc = 0$ for a nonzero state.

Example 7.2 · Bell-state entanglement

For $|\Phi^+\rangle$, $C = 2^{-1/2}I$ and $\det C = 1/2 \neq 0$; therefore the state is entangled.

Pure-state criterion

The determinant criterion applies to pure two-qubit state vectors. Mixed-state separability requires a convex decomposition and cannot be tested with $ad - bc$.

7.3 Density matrices

A pure state has density operator $\rho = |\psi\rangle\langle\psi|$. A mixed state is

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \quad p_k \geq 0, \quad \sum_k p_k = 1.$$

Every physical density matrix is Hermitian, positive, and has unit trace. It is pure exactly when $\text{tr}(\rho^2) = 1$; otherwise $\text{tr}(\rho^2) < 1$.

Example 7.3 · A classical mixture

The equal mixture of $|0\rangle$ and $|1\rangle$ is $\rho = I/2$. Its purity is $\text{tr}(\rho^2) = 1/2$. It predicts equal Z outcomes but differs from the pure superposition $|+\rangle$, which gives a definite X outcome.

Expectations generalize to $\langle A \rangle = \text{tr}(\rho A)$.

7.4 Partial trace

For a joint state ρ_{AB} , the reduced state of subsystem A is $\rho_A = \text{tr}_B(\rho_{AB})$. In a basis $\{|j\rangle_B\}$,

$$\rho_A = \sum_j (I_A \otimes \langle j|) \rho_{AB} (I_A \otimes |j\rangle).$$

Example 7.4 · Reduced state of a Bell pair

For $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$, $\rho_A = \rho_B = I/2$. A pure entangled joint state can therefore give maximally mixed local states.

7.5 Exercises

Exercise 7.1

Expand $(|0\rangle + i|1\rangle) \otimes |+\rangle/\sqrt{2}$ in the two-qubit basis.

Exercise 7.2

Use $ad - bc$ to classify $(|00\rangle + |01\rangle + |10\rangle + |11\rangle)/2$.

Exercise 7.3

Explain how $I/2$ and $|+\rangle\langle+|$ can agree for Z measurements but disagree for X measurements.

Exercise 7.4

Use NumPy to form a Bell density matrix, take a partial trace, and compute the purity.

References and further reading

Bengtsson, I., & Życzkowski, K. (2017). *Geometry of quantum states* (2nd ed.). Cambridge University Press.

Watrous, J. (2018). *The theory of quantum information*. Cambridge University Press.

8 Quantum gates and circuits

Learning objectives

- Interpret gates as unitary operators.
- Apply controlled gates and build an entangling circuit.
- Track operator order consistently in algebra and diagrams.

8.1 Single-qubit gates

Quantum gates are unitary and therefore reversible. Alongside X , Y , and Z , central single-qubit gates include

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

Example 8.1 • Hadamard interference

$H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$. Since $H^2 = I$, applying H again to $|+\rangle$ returns $|0\rangle$; the second gate recombines amplitudes coherently.

Example 8.2 • Phase gate

$S|+\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$. The Z probabilities remain equal, while the relative phase changes measurements in the X or Y basis.

8.2 Controlled gates

The controlled-NOT gate flips the target when the control is 1:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Example 8.3 · Preparing a Bell state

The circuit starts in $|00\rangle$, applies H to the first qubit, and then applies CNOT:

$$|00\rangle \xrightarrow{H \otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

The final state is entangled.

8.3 Order of operations

In matrix notation the rightmost operator acts first. A circuit that applies U_1 and then U_2 produces $U_2 U_1 |\psi\rangle$. Operators on distinct qubits, such as $A \otimes I$ and $I \otimes B$, commute; arbitrary gates do not.

Example 8.4 · Order matters

$HZ|0\rangle = H|0\rangle = |+\rangle$, whereas $ZH|0\rangle = Z|+\rangle = |-\rangle$. Therefore $HZ \neq ZH$.

Diagram order and matrix order

Circuit diagrams follow the left-to-right reading order, while the rightmost factor in an operator product acts first. Write an intermediate state after each layer when translating a circuit into algebra.

8.4 Exercises

Exercise 8.1

Compute HXH and identify the resulting Pauli operator.

Exercise 8.2

Apply CNOT to each computational basis state and verify that the matrix is unitary.

Exercise 8.3

Design a circuit that prepares $(|01\rangle + |10\rangle)/\sqrt{2}$ from $|00\rangle$.

Exercise 8.4

Simulate the Bell circuit with NumPy and verify the final probabilities and reduced states.

References and further reading

- Kaye, P., Laflamme, R., & Mosca, M. (2007). *An introduction to quantum computing*. Oxford University Press.
- Nielsen, M. A., & Chuang, I. L. (2010). *Quantum computation and quantum information*. Cambridge University Press.

PART III

Hamiltonians and optimization models

9 The classical Hamiltonian

Learning objectives

- Construct a Hamiltonian from a Lagrangian by a Legendre transform.
- Use Hamilton's equations to derive motion.
- Interpret Poisson brackets and conserved quantities.

9.1 The Legendre transform and the Hamiltonian

For generalized coordinates q_i and velocities $\dot{q}_i$, the Lagrangian is $L(q, \dot{q}, t)$. The conjugate momenta are

$$p_i = \frac{\partial L}{\partial \dot{q}_i}.$$

When the velocity-momentum relation can be inverted, the Legendre transform defines

$$H(q, p, t) = \sum_i p_i \dot{q}_i - L(q, \dot{q}, t).$$

Hamilton's equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

For standard mechanical systems, $H = T + V$ is the total energy. More generally, its defining role is to generate time evolution on phase space.

9.2 Example 1: free particle

For $H = p^2/(2m)$,

$$\dot{x} = \frac{p}{m}, \quad \dot{p} = 0.$$

Thus $p(t) = p_0$ and $x(t) = x_0 + p_0 t/m$.

9.3 Example 2: harmonic oscillator

For

$$H(x, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2,$$

Hamilton's equations give $\dot{x} = p/m$ and $\dot{p} = -m\omega^2 x$. Therefore $\ddot{x} + \omega^2 x = 0$, with sinusoidal solutions and conserved energy.

9.4 Poisson brackets

For functions $f(q, p, t)$ and $g(q, p, t)$,

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right).$$

The time evolution of f is

$$\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t}.$$

If f has no explicit time dependence and $\{f, H\} = 0$, then f is conserved.

Example 9.1 • Canonical variables

$\{q_i, p_j\} = \delta_{ij}$, while $\{q_i, q_j\} = \{p_i, p_j\} = 0$. These relations anticipate the canonical commutators of quantum mechanics.

9.5 Example 3: simple pendulum

For angular coordinate θ and moment of inertia $m\ell^2$,

$$H(\theta, p_\theta) = \frac{p_\theta^2}{2m\ell^2} + mg\ell(1 - \cos \theta).$$

Hamilton's equations yield $\ddot{\theta} + (g/\ell) \sin \theta = 0$. For small angles, $\sin \theta \approx \theta$ and the motion becomes harmonic with $\omega = \sqrt{g/\ell}$.

9.6 Example 4: central motion

For a particle in a central potential $V(r)$,

$$H = \frac{p_r^2}{2m} + \frac{L^2}{2mr^2} + V(r).$$

Rotational symmetry conserves the angular momentum L . The radial motion is equivalent to one-dimensional motion in the effective potential $V_{\text{eff}}(r) = L^2/(2mr^2) + V(r)$.

Bridge to quantum mechanics

The Poisson bracket generates classical evolution; the commutator generates quantum evolution. The correspondence $\{f, g\} \leftrightarrow [F, G]/(i\hbar)$ is a guide, not a universal algebraic substitution rule.

9.7 Exercises

Exercise 9.1

Derive the Hamiltonian of a particle with $L = m\dot{x}^2/2 - V(x)$.

Exercise 9.2

Use a Poisson bracket to prove that H is conserved when it has no explicit time dependence.

Exercise 9.3

Explain the physical meaning of the centrifugal term in V_{eff} .

Exercise 9.4

Integrate the harmonic-oscillator Hamilton equations numerically and monitor energy drift.

References and further reading

Goldstein, H., Poole, C., & Safko, J. (2002). *Classical mechanics* (3rd ed.). Addison-Wesley.

Taylor, J. R. (2005). *Classical mechanics*. University Science Books.

10 The quantum Hamiltonian

Learning objectives

- Use the Schrödinger equation and unitary time evolution.
- Expand a state in the energy eigenbasis.
- Identify conserved observables from commutators.

10.1 The Schrödinger equation

The Hamiltonian $H = H^\dagger$ represents energy and generates time evolution:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle.$$

For time-independent H ,

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad U(t) = e^{-iHt/\hbar}.$$

Hermiticity of H makes $U(t)$ unitary and preserves normalization.

10.2 Energy eigenstates

If $H|E_n\rangle = E_n|E_n\rangle$ and $|\psi(0)\rangle = \sum_n c_n |E_n\rangle$, then

$$|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle.$$

Each energy component acquires a phase. A single energy eigenstate changes only by a global phase and is therefore stationary in its measurement probabilities.

10.3 Example 1: spin one-half

For $H = (\hbar\omega/2)Z$ and $|\psi(0)\rangle = |+\rangle$,

$$|\psi(t)\rangle = \frac{e^{-i\omega t/2}|0\rangle + e^{i\omega t/2}|1\rangle}{\sqrt{2}}.$$

The probability of obtaining $+$ in an X measurement is $\cos^2(\omega t/2)$.

10.4 Example 2: coupled two-level system

For

$$H = \begin{pmatrix} E & g \\ g & E \end{pmatrix} = EI + gX,$$

the energies are $E \pm g$. Starting in $|0\rangle$,

$$|\psi(t)\rangle = e^{-iEt/\hbar} \left[\cos \frac{gt}{\hbar} |0\rangle - i \sin \frac{gt}{\hbar} |1\rangle \right].$$

The transition probability is $\sin^2(gt/\hbar)$.

10.5 Conserved quantities

For an observable A ,

$$\frac{d}{dt}\langle A \rangle = \frac{i}{\hbar}\langle [H, A] \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle.$$

If A has no explicit time dependence and $[H, A] = 0$, its expectation is conserved. If H is time independent, energy expectation is conserved.

10.6 Example 3: three energy levels

Let $H = \text{diag}(0, \hbar\omega, 2\hbar\omega)$ and $|\psi(0)\rangle = (|0\rangle + |1\rangle + |2\rangle)/\sqrt{3}$. Then

$$|\psi(t)\rangle = \frac{|0\rangle + e^{-i\omega t}|1\rangle + e^{-2i\omega t}|2\rangle}{\sqrt{3}}.$$

The energy probabilities remain 1/3, while observables that mix energy levels can oscillate because of relative phases.

10.7 Example 4: a commuting time-dependent Hamiltonian

If $H(t) = f(t)A$ for fixed Hermitian A , then $[H(t), H(t')] = 0$ and

$$U(t) = \exp \left[-\frac{iA}{\hbar} \int_0^t f(s) ds \right].$$

When Hamiltonians at different times do not commute, this simple exponential must be replaced by a time-ordered evolution.

Matrix exponential

Exponentiating matrix entries one by one is incorrect. Diagonalize the matrix, use a spectral expansion, or apply a verified matrix-exponential routine such as `scipy.linalg.expm`.

10.8 Exercises

Exercise 10.1

Find $U(t)$ for $H = \hbar\omega X/2$ and apply it to $|0\rangle$.

Exercise 10.2

Prove that $e^{-iHt/\hbar}$ is unitary when H is Hermitian.

Exercise 10.3

Explain why a stationary state may still have a time-dependent phase.

Exercise 10.4

Use SciPy to compare the spectral and direct matrix exponential for a random 3×3 Hermitian matrix.

References and further reading

- Griffiths, D. J., & Schroeter, D. F. (2018). *Introduction to quantum mechanics* (3rd ed.). Cambridge University Press.
- Sakurai, J. J., & Napolitano, J. (2021). *Modern quantum mechanics* (3rd ed.). Cambridge University Press.

11 Continuous systems and unbounded operators

Learning objectives

- Use position and momentum operators in wave mechanics.
- Solve standard energy-eigenvalue problems.
- Recognize why operator domains matter in infinite dimensions.

11.1 Position and momentum

In the position representation,

$$(\hat{x}\psi)(x) = x\psi(x), \quad (\hat{p}\psi)(x) = -i\hbar \frac{d\psi}{dx}.$$

They satisfy $[\hat{x}, \hat{p}] = i\hbar I$. For a particle of mass m in a potential $V(x)$,

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x).$$

11.2 Example 1: particle in a box

For $0 < x < L$ with $\psi(0) = \psi(L) = 0$,

$$\phi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, \quad E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = 1, 2, \dots$$

The boundary conditions discretize the energy.

11.3 Example 2: quantum harmonic oscillator

For $V(x) = m\omega^2 x^2/2$,

$$H = \hbar\omega \left(a^\dagger a + \frac{1}{2} \right), \quad E_n = \hbar\omega \left(n + \frac{1}{2} \right).$$

The nonzero ground-state energy reflects the uncertainty principle.

11.4 Example 3: free particle

For $V = 0$, plane waves e^{ikx} have momentum $p = \hbar k$ and energy $E = \hbar^2 k^2/(2m)$. A single plane wave is not square-integrable over the full line; physical localized states are wave packets formed from superpositions.

11.5 Example 4: particle on a ring

With $\psi(\varphi + 2\pi) = \psi(\varphi)$,

$$\phi_n(\varphi) = \frac{e^{in\varphi}}{\sqrt{2\pi}}, \quad E_n = \frac{\hbar^2 n^2}{2I}, \quad n \in \mathbb{Z}.$$

Periodicity quantizes angular momentum.

11.6 Hermitian, symmetric, and self-adjoint

In finite dimensions, Hermitian and self-adjoint are equivalent. In an infinite-dimensional Hilbert space, an operator includes both its action and its domain. A symmetric operator satisfies $\langle \phi | A\psi \rangle = \langle A\phi | \psi \rangle$ on its domain, but it is self-adjoint only when the adjoint has the same domain. Boundary conditions can therefore change the physical operator and its spectrum even when the differential expression looks identical.

Domains are physical information

The expression $-i\hbar d/dx$ does not by itself specify a momentum operator. The interval, regularity, and boundary conditions determine whether it is self-adjoint and which measurement outcomes are allowed.

11.7 Exercises

Exercise 11.1

Normalize the ground state of a particle in a box and compute $\langle x \rangle$.

Exercise 11.2

Show that $[x, -i\hbar d/dx]\psi = i\hbar\psi$ on a suitable domain.

Exercise 11.3

Explain why a plane wave is a useful generalized eigenfunction but not a normalizable state on $\mathbb{R}$.

Exercise 11.4

Discretize $-d^2/dx^2$ with finite differences and compare its first four eigenvalues with the particle-in-a-box formula.

References and further reading

Hall, B. C. (2013). *Quantum theory for mathematicians*. Springer.

Reed, M., & Simon, B. (1980). *Methods of modern mathematical physics I: Functional analysis*. Academic Press.

12 Ising Hamiltonians, QUBO, and quantum annealing

Learning objectives

- Express binary optimization with Ising and QUBO objectives.
- Convert between spin and binary variables.
- Interpret the driver and problem Hamiltonians in quantum annealing.

12.1 The Ising model

For spins $s_i \in \{-1, +1\}$, the classical Ising energy is

$$E(s) = \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j.$$

The coefficients h_i are local biases and J_{ij} are couplings. Optimization asks for a spin assignment that minimizes $E(s)$.

The corresponding problem Hamiltonian is diagonal in the computational basis:

$$H_P = \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j.$$

Example 12.1 · Two coupled spins

For $E = -s_1 s_2$, aligned spins have energy -1 and anti-aligned spins have energy $+1$. The two ground states are $(+1, +1)$ and $(-1, -1)$.

12.2 QUBO formulation

A quadratic unconstrained binary optimization problem uses $x_i \in \{0, 1\}$ and objective

$$Q(x) = \sum_i Q_{ii} x_i + \sum_{i < j} Q_{ij} x_i x_j.$$

The transformations

$$s_i = 2x_i - 1, \quad x_i = \frac{1 + s_i}{2}$$

convert between QUBO and Ising forms, up to an additive constant that does not change the minimizer.

Example 12.2 · Encoding a constraint

To encourage exactly one of two bits to be 1, add

$$\lambda(x_1 + x_2 - 1)^2 = \lambda(1 - x_1 - x_2 + 2x_1 x_2),$$

using $x_i^2 = x_i$ for binary variables. The penalty vanishes only for $(1, 0)$ and $(0, 1)$.

Example 12.3 · Enumerating a small QUBO

For $Q(x) = -x_1 - x_2 + 2x_1 x_2$, the values at $(0, 0), (0, 1), (1, 0), (1, 1)$ are $0, -1, -1, 0$. Exhaustive enumeration is a useful validation method for small formulations.

12.3 Quantum annealing

Quantum annealing uses a time-dependent Hamiltonian

$$H(s) = A(s)H_D + B(s)H_P, \quad 0 \leq s \leq 1,$$

where H_D is a driver with an easily prepared ground state and H_P encodes the objective. A common driver is $H_D = -\sum_i X_i$. Ideally, slow evolution keeps the state near the instantaneous ground state. Practical outcomes depend on the minimum spectral gap, temperature, noise, controls, embedding, and total anneal time.

Example 12.4 · What is actually sampled

The device returns a collection of bit strings and energies. The minimum sample found is evidence about the optimization run, not a proof that the global minimum was obtained. Small instances should be checked by exact enumeration.

Penalty coefficients

A penalty that is too small permits infeasible solutions; one that is too large can compress the useful objective scale and amplify control errors. Derive a valid lower bound, then test sensitivity instead of choosing a large number without analysis.

12.4 Exercises

Exercise 12.1

Enumerate all energies of $E = s_1 + s_2 - s_1 s_2$ and identify the ground states.

Exercise 12.2

Convert $Q(x) = 3x_1 - 2x_2 + 4x_1x_2$ into an Ising objective.

Exercise 12.3

Explain why an additive constant can be dropped during optimization but should be restored when comparing absolute energy values.

Exercise 12.4

Write a Python enumerator for a three-variable QUBO and verify every reported minimum.

References and further reading

Lucas, A. (2014). Ising formulations of many NP problems. *Frontiers in Physics*, 2, 5.

McGeoch, C. C. (2014). *Adiabatic quantum computation and quantum annealing*. Morgan & Claypool.

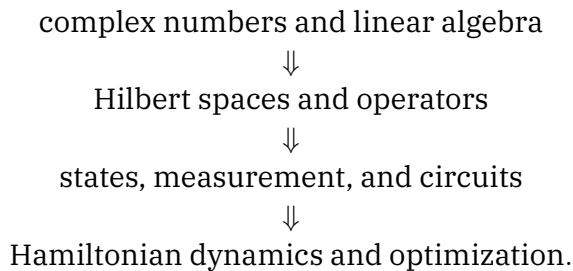
PART IV

Synthesis and computational practice

13 Synthesis, common errors, and study path

13.1 Concept map

The book's dependency chain can be summarized as follows:



Each arrow represents a real dependency. Complex amplitudes require conjugation; probabilities require inner products; observables require the spectral theorem; composite registers require tensor products; and dynamics requires unitary evolution generated by a self-adjoint Hamiltonian.

13.2 Classical and quantum correspondence

ROLE	CLASSICAL MECHANICS	QUANTUM MECHANICS
State	Point (q, p) in phase space	Ray or density operator on $\mathcal{H}$
Observable	Function $f(q, p)$	Self-adjoint operator A
Dynamics	$\dot{f} = \{f, H\} + \partial_t f$	$\langle \dot{A} \rangle = \frac{i}{\hbar} \langle [H, A] \rangle + \langle \partial_t A \rangle$
Composition	Cartesian product	Tensor product
Uncertainty	Statistical or instrumental	Intrinsic plus statistical and instrumental

13.3 Common errors

Using a transpose instead of an adjoint

Complex inner products require conjugation. Replace v^T by $v^\dagger$.

Confusing amplitudes and probabilities

Add amplitudes for indistinguishable alternatives, then square the modulus. Adding probabilities first destroys interference.

Ignoring basis order

State explicitly whether two-qubit coordinates use $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ and which qubit is the control.

Treating an estimate as an exact expectation

Finite counts fluctuate. Record the number of shots, uncertainty, seed, and sampling procedure.

Applying finite-dimensional intuition without a domain

Differential operators require boundary conditions and domains. Formal integration by parts alone does not establish self-adjointness.

13.4 Recommended path

Readers preparing for a first quantum-computing course should master Chapters 1 through 8, complete Laboratories 1 through 4, and then study Chapters 9 through 12. Readers interested primarily in optimization may begin with Chapters 1, 2, 5, and 6, continue with Chapters 10 and 12, and review density matrices and sampling before interpreting hardware results.

The best test of mastery is the ability to move among four representations of the same statement: ordinary language, algebra, a geometric picture, and a reproducible numerical check.

14 Reproducible computational laboratories

14.1 Environment and reproducibility

The laboratories require Python 3, NumPy, SciPy, and Matplotlib. Record the package versions, random seed, input data, and commands used. A notebook is reproducible only when another reader can recreate both the numerical result and the figure from the recorded environment.

```
python -m venv .venv
source .venv/bin/activate
python -m pip install numpy scipy matplotlib
python -m pip freeze > requirements-lock.txt
```

Lab 14.1 · Adjoint, spectrum, and tensor product

Construct a random complex matrix M and set $A = (M + M^\dagger)/2$. Verify $A = A^\dagger$, compute its eigenpairs with `numpy.linalg.eigh`, confirm $Av_k \approx \lambda_k v_k$, and build $A \otimes I_2$ with `numpy.kron`. Report the largest numerical residual.

```
import numpy as np
rng = np.random.default_rng(2026)
M = rng.normal(size=(3, 3)) + 1j*rng.normal(size=(3, 3))
A = (M + M.conj().T) / 2
w, V = np.linalg.eigh(A)
residual = np.linalg.norm(A @ V - V @ np.diag(w))
print(residual, np.linalg.norm(A - A.conj().T))
```

Lab 14.2 · Bell state, density matrix, and partial trace

Build $|\Phi^+\rangle$, form $\rho = |\Phi^+\rangle\langle\Phi^+|$, reshape ρ as a tensor with four subsystem indices, trace out each qubit, and verify that both reduced states equal $I/2$. Report trace, Hermiticity error, and purity.

Lab 14.3 · Probabilities and finite counts

Sample a Bernoulli distribution with $p = 0.3$ for $N = 10^2, 10^3, 10^4, 10^5$. Repeat the experiment at least 100 times for each N . Plot the root-mean-square estimation error against N on logarithmic axes and verify a slope near $-1/2$.

Lab 14.4 · Bloch sphere and ideal tomography

Generate normalized qubits and compute $r_x = \text{tr}(\rho X)$, $r_y = \text{tr}(\rho Y)$, and $r_z = \text{tr}(\rho Z)$. Reconstruct $\rho = (I + \mathbf{r} \cdot \boldsymbol{\sigma})/2$ and compare it with the original. Include pure and mixed states.

Lab 14.5 · Evolution by a matrix exponential

Choose a Hermitian 2×2 Hamiltonian and compute $U(t) = \text{scipy.linalg.expm}(-iHt/\hbar)$. Verify unitarity and norm preservation for a time grid. Compare the numerical result with a spectral decomposition.

Lab 14.6 · Discretization and QUBO verification

Discretize the particle-in-a-box Hamiltonian with a centered finite-difference Laplacian and compare its low eigenvalues with the analytic spectrum. Then write a separate exhaustive enumerator for a three-variable QUBO. The report must state the discretization error and list every minimizing bit string.

Evidence before interpretation

Every laboratory must include an analytic expectation, a numerical result, a residual or uncertainty, and a brief explanation of any discrepancy. A plot without a quantitative check is an illustration, not validation.

A Reasoned solutions to the exercises

The solutions emphasize method, verification, and common errors. Computational answers should be reproduced independently with the versions and random seeds recorded in the laboratory report.

A.1 Chapter 1: mathematical prerequisites

Exercise 1.1

Let $u = (1, i)^T$ and $v = (i, 1)^T$. Then $u^\dagger v = (1, -i)(i, 1)^T = i - i = 0$, so the vectors are orthogonal. The common error is to use u^T and obtain $2i$.

Exercise 1.2

Set $v_1 = (1, 1)^T$ and $v_2 = (1, i)^T$. First, $u_1 = v_1/\sqrt{2}$. Then

$$w_2 = v_2 - \langle u_1 | v_2 \rangle u_1 = \frac{1-i}{2}(1, -1)^T,$$

and a convenient normalized result is $u_2 = (1-i)(1, -1)^T/2$. Multiplying u_2 by a global phase gives an equivalent orthonormal basis.

Exercise 1.3

Coordinates are coefficients relative to a chosen basis. The same vector has column $(3, 1)^T$ in the canonical basis and $(2, 1)^T$ in the basis $\{(1, 1)^T, (1, -1)^T\}$. Confusing the vector with one coordinate list hides the transformation law under basis changes.

Exercise 1.4

`numpy.linalg.eigh` returns eigenvalues 1 and 3 and orthonormal eigenvectors up to sign or phase. Verify residuals with $\|Av - \lambda v\|$. The Kronecker product has shape 4×4 . A result that depends on exact eigenvector signs is overconstrained because eigenvectors are not unique under phase multiplication.

A.2 Chapter 2: complex numbers

Exercise 2.1

For $z = -1 + i\sqrt{3}$, $r = 2$ and $\arg z = 2\pi/3$, so $z = 2e^{2\pi i/3}$. Angles differing by $2\pi k$ are equivalent.

Exercise 2.2

For $v = (1 - i, 2)^T$, $v^\dagger v = 2 + 4 = 6$, hence $\psi = (1 - i, 2)^T/\sqrt{6}$. The norm check must use conjugation.

Exercise 2.3

If $|0\rangle - |1\rangle = e^{i\gamma}(|0\rangle + |1\rangle)$, comparison of the first coefficient requires $e^{i\gamma} = 1$, while comparison of the second requires $e^{i\gamma} = -1$. No single phase satisfies both; the relative phase changed.

Exercise 2.4

Generate $z_k = e^{ik\pi/2}$ and plot their real and imaginary parts. Numerical z_k^4 should differ from 1 only by floating-point roundoff. Use a tolerance, not exact binary equality.

A.3 Chapter 3: probability

Exercise 3.1

$\mathbb{E}[X] = 0(0.7) + 1(0.3) = 0.3$ and $\text{Var}(X) = 0.3 - 0.3^2 = 0.21$.

Exercise 3.2

$\hat{p} = 0.24$ and $SE = \sqrt{0.24(0.76)/1000} \approx 0.0135$. The approximate 95 percent interval is $0.24 \pm 1.96(0.0135)$, or about $[0.214, 0.266]$. Do not report more precision than the sample supports.

Exercise 3.3

Indistinguishable alternatives combine as $|a_1 + a_2|^2$, which includes $2 \operatorname{Re}(a_1^* a_2)$. Adding $|a_1|^2 + |a_2|^2$ discards this interference term.

Exercise 3.4

Across repeated simulations, the root-mean-square error should be close to $\sqrt{p(1-p)/N}$. A log-log fit should have slope near $-1/2$. One simulation per N is inadequate because its random fluctuation can obscure the scaling.

A.4 Chapter 4: Hilbert spaces**Exercise 4.1**

The vector has coefficients $1/\sqrt{3}$, $i/\sqrt{3}$, $-1/\sqrt{3}$, and 0. Their squared moduli sum to 1, proving normalization.

Exercise 4.2

$P = |+\rangle\langle+| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ is real symmetric, so $P^\dagger = P$, and direct multiplication gives $P^2 = P$.

Exercise 4.3

The operator $\sum_n |n\rangle\langle n|$ reproduces only the component in the span of the listed vectors. It equals I only when that span is the whole space.

Exercise 4.4

Reshape the Bell coefficients into $C = 2^{-1/2}I$. Since its matrix rank is 2, the state cannot factor into two vectors. Use a tolerance with `matrix_rank`; a floating-point determinant alone can be misleading near rank deficiency.

A.5 Chapter 5: operators

Exercise 5.1

The off-diagonal entries $1 + i$ and $1 - i$ are conjugates and the diagonal is real, so the matrix equals its adjoint and is Hermitian.

Exercise 5.2

If $Pv = \lambda v$, then $P^2v = \lambda^2v$, but $P^2 = P$ also gives $P^2v = \lambda v$. Therefore $\lambda(\lambda - 1) = 0$ and $\lambda \in \{0, 1\}$.

Exercise 5.3

X and Y are Hermitian, but $XY = iZ$ is anti-Hermitian rather than Hermitian. The product of Hermitian operators is Hermitian only when the factors commute.

Exercise 5.4

Numerically verify $XY - YX = 2iZ$, $YZ - ZY = 2iX$, and $ZX - XZ = 2iY$ with `numpy.allclose`. Reversing a commutator reverses its sign.

A.6 Chapter 6: measurement

Exercise 6.1

The state has equal Z probabilities, so $\langle Z \rangle = 0$. Since $Z^2 = I$, $(\Delta Z)^2 = 1 - 0 = 1$ and $\Delta Z = 1$.

Exercise 6.2

Here $\theta/2 = \pi/8$ and $\phi = 0$, so $\mathbf{r} = (\sin(\pi/4), 0, \cos(\pi/4)) = (1/\sqrt{2}, 0, 1/\sqrt{2})$.

Exercise 6.3

After outcome a , the ideal state lies in the eigenspace projected by P_a . A repeated measurement applies the same projector, so it returns a with probability one. This statement assumes no intervening evolution or noise.

Exercise 6.4

The theoretical Z expectation is $0.3 - 0.7 = -0.4$. The simulated sample mean should lie near this value with standard error approximately $\sqrt{1 - (-0.4)^2}/\sqrt{10000} \approx 0.0092$.

A.7 Chapter 7: multiple qubits and density matrices

Exercise 7.1

The normalized first factor is $(|0\rangle + i|1\rangle)/\sqrt{2}$. Tensoring with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ gives $(|00\rangle + |01\rangle + i|10\rangle + i|11\rangle)/2$.

Exercise 7.2

All coefficients equal $1/2$, so $ad - bc = 1/4 - 1/4 = 0$. The state is $|+\rangle \otimes |+\rangle$ and is separable.

Exercise 7.3

Both states give probabilities $1/2, 1/2$ for Z . For X , $|+\rangle$ yields $+1$ with certainty, while $I/2$ gives ± 1 equally. Agreement for one measurement basis does not identify a quantum state.

Exercise 7.4

The Bell density matrix has trace one, is Hermitian, and has purity one. Tracing either subsystem yields $I/2$ with purity $1/2$. Be explicit about the tensor index order when reshaping.

A.8 Chapter 8: gates and circuits

Exercise 8.1

Direct multiplication gives $HXH = Z$. This expresses a basis change between the X and Z observables.

Exercise 8.2

CNOT maps $|00\rangle \rightarrow |00\rangle$, $|01\rangle \rightarrow |01\rangle$, $|10\rangle \rightarrow |11\rangle$, and $|11\rangle \rightarrow |10\rangle$. Its columns are an orthonormal permutation of the basis, hence it is unitary.

Exercise 8.3

Apply X to the second qubit, H to the first, and CNOT with the first as control. The sequence produces $|01\rangle \rightarrow (|01\rangle + |11\rangle)/\sqrt{2} \rightarrow (|01\rangle + |10\rangle)/\sqrt{2}$.

Exercise 8.4

The state vector should be $(1, 0, 0, 1)^T/\sqrt{2}$, with probabilities $(1/2, 0, 0, 1/2)$. Each reduced density matrix should equal $I/2$. A reversed qubit order can appear correct for this symmetric state, so also test an asymmetric basis input.

A.9 Chapter 9: classical Hamiltonian mechanics

Exercise 9.1

$p = \partial L / \partial \dot{x} = m\dot{x}$, so $\dot{x} = p/m$. Then $H = p\dot{x} - L = p^2/(2m) + V(x)$.

Exercise 9.2

$dH/dt = \{H, H\} + \partial H/\partial t$. Antisymmetry gives $\{H, H\} = 0$; therefore a time-independent Hamiltonian is conserved.

Exercise 9.3

$L^2/(2mr^2)$ grows as r decreases and acts as a centrifugal barrier in the radial problem. It is kinetic energy associated with angular motion, expressed as part of an effective one-dimensional potential.

Exercise 9.4

A symplectic method should exhibit bounded energy error, while a naive forward Euler method usually shows systematic drift. Compare methods at several step sizes before attributing drift to physics.

A.10 Chapter 10: quantum Hamiltonians

Exercise 10.1

Since $X^2 = I$,

$$U(t) = \cos \frac{\omega t}{2} I - i \sin \frac{\omega t}{2} X.$$

Thus $U(t)|0\rangle = \cos(\omega t/2)|0\rangle - i \sin(\omega t/2)|1\rangle$.

Exercise 10.2

$U^\dagger = e^{iHt/\hbar}$ because $H^\dagger = H$. Both exponentials are functions of the same matrix, so they commute and $U^\dagger U = I$.

Exercise 10.3

An energy eigenstate acquires $e^{-iE_n t/\hbar}$, a global phase. All Born probabilities and expectations are unchanged even though the representing vector depends on time.

Exercise 10.4

The two methods should agree within numerical tolerance. Check both $\|U_{\text{expm}} - U_{\text{spectral}}\|$ and unitarity. Use `eigh`, not a general eigensolver, to exploit Hermiticity.

A.11 Chapter 11: continuous systems

Exercise 11.1

$\phi_1 = \sqrt{2/L} \sin(\pi x/L)$ is normalized and symmetric about $L/2$ in probability density, so $\langle x \rangle = L/2$. Direct integration confirms it.

Exercise 11.2

$x(-i\hbar\psi') - [-i\hbar(x\psi)'] = i\hbar\psi$. The calculation assumes that ψ lies in a domain where the derivatives exist.

Exercise 11.3

A plane wave has constant modulus, so its integral over $\mathbb{R}$ diverges. It is useful as a generalized momentum eigenfunction and as a component of normalizable wave packets.

Exercise 11.4

With grid spacing h , the negative second derivative has diagonal $2/h^2$ and off-diagonal $-1/h^2$. Its low eigenvalues converge to $(n\pi/L)^2$ as $h \rightarrow 0$. Report convergence with grid refinement rather than one discretization.

A.12 Chapter 12: Ising and QUBO

Exercise 12.1

For $(s_1, s_2) = (1, 1), (1, -1), (-1, 1), (-1, -1)$, the energies are 1, 1, 1, -3. The unique ground state is $(-1, -1)$.

Exercise 12.2

Substitute $x_i = (1 + s_i)/2$ and collect terms:

$$Q = \text{constant} + \frac{5}{2}s_1 + 0s_2 + s_1s_2,$$

with constant $3/2$. Check the four assignments in both encodings to catch a sign or convention error.

Exercise 12.3

Adding a constant shifts every energy equally, leaving their ordering and minimizers unchanged. Absolute comparisons with another formulation require the constant to avoid reporting inconsistent objective values.

Exercise 12.4

Enumerate $2^3 = 8$ bit strings, evaluate the objective directly, and retain all strings tied at the minimum. Do not assume the minimizer is unique, and test the program against a hand-calculated instance.

B

Essential notation

SYMBOL	MEANING
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}$	Natural, integer, and rational numbers.
$\mathbb{R}, \mathbb{C}$	Real and complex number fields.
$\mathbb{F}$	A generic scalar field, normally $\mathbb{R}$ or $\mathbb{C}$.
$V, \mathcal{H}$	Vector space and Hilbert space.
$A^T, A^*, A^\dagger$	Transpose, entrywise conjugate, and adjoint.
$ \psi\rangle, \langle\psi $	State vector and its dual.
$\langle\phi \psi\rangle$	Inner product.
$ \psi\rangle\langle\phi $	Outer-product operator.
$\ \psi\ , z $	Vector norm and scalar modulus.
$A \otimes B$	Tensor product.
$[A, B]$	Commutator $AB - BA$.
$\{f, g\}$	Classical Poisson bracket.
ρ	Density operator.
tr_B	Partial trace over subsystem B .
X, Y, Z	Pauli operators.
H	Hamiltonian; in a circuit context, also the Hadamard gate when clear.
$\hbar$	Reduced Planck constant.
$\langle A \rangle$	Expectation value of observable A .
ΔA	Standard deviation of observable A .
h_i, J_{ij}	Ising local biases and couplings.

SYMBOL	MEANING
Q_{ij}	QUBO coefficients.

C Proposed teaching organization

C.1 Learning outcomes

By the end of the sequence, students should be able to represent quantum states and observables correctly, calculate measurement statistics, analyze small circuits and Hamiltonians, formulate elementary Ising and QUBO models, and verify claims with reproducible numerical experiments.

C.2 Time accounting

One contact hour means a scheduled hour with an instructor. Independent work includes reading, exercises, laboratory preparation, and reports. The plans below assume two hours of independent work for each contact hour.

C.3 Option I: one complete course

C.3.1 Profile and duration

The compact option runs for 16 weeks with four contact hours per week. It contains 64 contact hours and approximately 128 total learning hours. Students should enter with high-school algebra, elementary calculus, and basic Python.

C.3.2 Weekly plan

WEEK	TOPICS	PRACTICE
1	Sets, functions, number systems, scalars, and vectors	Diagnostic exercises
2	Bases, matrices, linear maps, kernel, image, and rank	Matrix calculations
3	Inner products, Gram-Schmidt, eigenvalues, and tensor products	Lab 1 begins
4	Complex numbers, phases, and adjoints	Lab 1 report
5	Probability, Bayes' rule, amplitudes, and finite counts	Lab 3 begins
6	Hilbert spaces, Dirac notation, and projectors	Problem set
7	Operators and the spectral theorem	Midterm review
8	Measurement, expectation, uncertainty, and Bloch vectors	Midterm examination
9	Multiple qubits, entanglement, and density matrices	Lab 2
10	Gates, circuits, and Bell-state preparation	Circuit design
11	Classical Hamiltonian mechanics and Poisson brackets	Numerical oscillator
12	Quantum Hamiltonians and unitary evolution	Lab 5
13	Continuous systems and operator domains	Lab 6 begins
14	Ising, QUBO, penalties, and annealing	QUBO validation
15	Integrated project workshop	Project evidence
16	Presentations and synthesis	Final assessment

C.3.3 Suggested assessment

Exercises account for 20 percent, laboratories for 30 percent, the midterm for 20 percent, and an integrated final project for 30 percent. At least half of the laboratory grade should measure reproducibility and interpretation rather than successful code execution alone.

C.4 Option II: a three-course sequence

The extended option comprises three 12-week courses, each with four contact hours per week. The sequence provides 144 contact hours and approximately 288 total learning hours.

C.4.1 Course I: mathematical and computational language

Chapters 1 through 5, Laboratories 1 and 3, and structured work in Python. Students finish by implementing linear-algebra tests with explicit numerical tolerances.

C.4.2 Course II: states, measurement, and circuits

Chapters 6 through 8, Laboratories 2 and 4, sampling experiments, tomography, and a small circuit-simulation project.

C.4.3 Course III: Hamiltonians and optimization models

Chapters 9 through 12, Laboratories 5 and 6, numerical time evolution, finite-difference spectra, and an Ising or QUBO capstone whose small instances are verified exhaustively.

C.5 Choosing an option

The one-course format suits a prerequisite or intensive introduction. The three-course sequence suits an undergraduate concentration or graduate bridge where students need time to develop proofs, software discipline, and physical interpretation. In either format, remove topics only after preserving their dependencies; for example, circuits should not precede tensor products and measurement statistics should not omit finite sampling.

Recommended bibliography for further study

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Information about the alliance is available at <https://www.cybercolombia.org/quantum/>.

Join the Quantum Colombia Alliance

If you would like to participate in the Quantum Colombia Alliance, write to quantum@cybercolombia.org.

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